CHAPTER 4: Compounds — Formulae and Naming

(a) COMPOUNDS CONTAINING ONLY NON-METALS

(i) *Most compounds containing only non-metals exist as molecules* in which the atoms are bound together by sharing electrons (covalent bonding).
Being molecular, the *formula* of each of these compounds indicates the composition of the molecules.

(ii) *A few compounds containing only non-metals, such as silicon carbide and silicon dioxide, exist as covalent network lattices.* Thus the *formula* of each of these compounds is just the simplest ratio of atoms in the lattice e.g. SiC and SiO_2.

Naming compounds of two non-metals.

Unless common names are used, the name of a non-metal compound also tells us the composition of the molecules, or the simplest ratio of atoms in the lattice; that is, *the formula is self evident from the name.*

Rules for Naming

1. *The name of the element closer to the bottom, left-hand corner of the periodic table is given first.*
2. *The name of the second element is altered to end with the suffix "-ide".*
3. *Prefixes are used to indicate the number of atoms of each element, except for the case of only one atom of the first named element.*

NUMBER OF ATOMS	*PREFIX*
1	*mono-*
2	*di-*
3	*tri-*
4	*tetra-*
5	*penta-*
6	*hexa-*
7	*hepta-*
8	*octa-*
9	*nona-*
10	*deca-*

The charge on the ions of an element is characteristic of that element. For example: sodium always forms Na^+ ions by loss of one electron; oxygen always forms O^{2-} ions by gain of two electrons.

Polyatomic ions consist of more than one type of atom bound together and having -

(i) less electrons than protons if positive.
(ii) more electrons than protons if negative.

Polyatomic ions are often referred to as radicals.

NAMES AND FORMULAE FOR SOME COMMON IONS AND POLYATOMIC IONS

+1 charge	*−1 charge*
lithium, Li^+ sodium, Na^+ potassium, K^+ copper (I) or cuprous, Cu^+ silver, Ag^+	fluoride, F^- chloride, Cl^- bromide, Br^- iodide, I^- hydride, H^-
ammonium, NH_4^+ hydrogen, $H^+_{(aq)}$	hydroxide, OH^- nitrite, NO_2^- nitrate, NO_3^- acetate, CH_3COO^- hydrogencarbonate or bicarbonate, HCO_3^- hydrogensulfate or bisulfate, HSO_4^- chlorate, ClO_3^-
+2 charge	*−2 charge*
manganese Mn^{2+} magnesium, Mg^{2+} calcium, Ca^{2+} barium, Ba^{2+} zinc, Zn^{2+} copper (II) or cupric, Cu^{2+} mercury (II) or mercuric, Hg^{2+} iron (II) or ferrous, Fe^{2+} tin (II) or stannous, Sn^{2+} lead (II) or plumbous, Pb^{2+} nickel(II), Ni^{2+}	oxide, O^{2-} sulfide, S^{2-}
	carbonate, CO_3^{2-} sulfate, SO_4^{2-} sulfite, SO_3^{2-}

+3 charge	*−3 charge*
aluminium, Al^{3+} iron (III) or ferric, Fe^{3+} chromium, Cr^{3+}	nitride, N^{3-}
	phosphate, PO_4^{3-}
+4 charge	
tin (IV) or stannic, Sn^{4+} lead (IV) or plumbic, Pb^{4+}	

THESE NAMES AND FORMULAE MUST BE MEMORISED

EXAMPLE 2: An ion formula and an ion name are given.

(i) Cl^-

(ii) aluminium ion

For each ion calculate the number of protons, the number of electrons gained or lost in forming the ion and the number of electrons in the ion.

(i) Cl^-

$Z = 17$ = number of protons

charge = -1 ∴ 1 more electron than protons i.e. 1 electron gained

∴ number of electrons = 18.

(ii) ion formula: Al^{3+}

$Z = 13$ = number of protons

charge = $+3$ ∴ 3 less electrons than protons i.e. 3 electrons lost

∴ number of electrons = 10.

EXERCISE 6 For each ion write the name, calculate the number of protons, the number of electrons gained or lost in forming the ion, and the number of electrons in the ion:

1. O^{2-}
2. Ba^{2+}
3. I^-
4. Fe^{2+}
5. N^{3-}

EXAMPLE 6: (a) Write the formula of the compound

(i) carbon monoxide *(ii) nitrogen dioxide*

(i) carbon monoxide: CO (ii) nitrogen dioxide: NO_2

(b) Name the compound with the formula

(i) CO_2 *(ii)* N_2O_4

(i) CO_2: carbon dioxide (ii) N_2O_4: dinitrogen tetroxide

Some common names and associated formulae that should be memorised are:

COMMON NAME	*FORMULA*
water	H_2O
ammonia	NH_3
hydrogen sulfide	H_2S
hydrogen peroxide	H_2O_2
methane	CH_4

Aqueous Solutions of Covalent Molecular Compounds

Aqueous solutions of a *few* covalent molecular compounds such as hydrogen chloride, HCl, hydrogen nitrate, HNO_3, and hydrogen sulfate, H_2SO_4, contain no molecules. In solution all molecules of these compounds are ionised and the solutions are known as hydrochloric acid, nitric acid and sulfuric acid, respectively. They contain the hydrogen ion in solution, $H^+(aq)$.

Aqueous solutions of some covalent molecular compounds contain both molecules and ions. For example, phosphoric acid, H_3PO_4, acetic acid, CH_3COOH, sulfurous acid, H_2SO_3, and carbonic acid, H_2CO_3.

Aqueous solutions of *most* covalent molecular compounds contain only molecules.

EXERCISE 19 Write formulae for the following compounds containing only non-metals.

1. hydrogen chloride
2. sulfur dioxide
3. phosphorus tribromide
4. carbon tetrachloride
5. tetraphosphorus decaoxide

EXCERCISE 20 Write names for the following compounds containing only non-metals.

1. SO_3 2. PCl_5 3. CF_4 4. NBr_3 5. S_2Cl_4

EXERCISE 21 Write either names or formulae for the following compounds containing only non-metals.

1. OCl_2
2. nitrogen trifluoride
3. HBr
4. silicon dioxide
5. SF_6

(b) COMPOUNDS OF A METAL WITH NON-METALS

Compounds of a metal with non-metals exist in a continuous ionic lattice in which positive metal ions and negative non-metal ions occupy the positions in the lattice.

The overall charge on the lattice is zero.

IONIC BONDING is the electrical attraction between positive ions and negative ions. This is the strong bonding which extends throughout the lattice.

Being a continuous lattice structure *the formula of a compound of a metal with non-metals* is just the simplest ratio of the ions in the lattice. The ratio is such that the ionic compound is electrically neutral.

Ionic compounds also include ammonium compounds, where no metal is involved. In compounds such as these the ammonium ion is similar to a metal ion.

Naming Ionic Compounds

The name of the positive metal ion is followed by the name of the negative non-metal ion.

There is no need to indicate the ratio of different ions because this ratio is fixed by the charge on the ions.

Rules for Writing Ionic Formulae

1. *Write down the formulae for the ions or polyatomic ions involved, giving the positive ion first.*
2. *Transfer the charge numbers in cross-like fashion to the bottom right hand side of each formula.*

3. *Cancel these transferred numbers to their simplest ratio.*
4. *Write the formula of the compound, including brackets around polyatomic ions if more than one is involved.*

EXAMPLE 7: Write the formula of
(a) calcium oxide (b) aluminium sulfide

(a) calcium oxide

Rule 1 Ca^{2+} O^{2-}

Rule 2 $(Ca^{2+})_2(O^{2-})_2$

Rule 3 $(Ca^{2+})_1(O^{2-})_1$

Rule 4 CaO

(b) aluminium sulfide

Rule 1 Al^{3+} S^{2-}

Rule 2 $(Al^{3+})_2(S^{2-})_3$

Rule 3 Not required

Rule 4 Al_2S_3

EXERCISE 22 Write formulae for —

1. potassium iodide
2. lithium oxide
3. zinc sulfide
4. barium nitride
5. copper(II) bromide
6. lead(IV) oxide
7. aluminium oxide
8. magnesium oxide
9. zinc fluoride
10. silver chloride

EXERCISE 23 Write names for —

1. $BaCl_2$ 2. ZnO 3. KBr 4. Mg_3N_2 5. FeI_2

EXAMPLE 8: Write the formula of
(a) calcium hydroxide (b) ammonium carbonate

(a) calcium hydroxide

Rule 1 Ca^{2+} OH^-

Rule 2 $(Ca^{2+})_1(OH^-)_2$

Rule 3 Not required

Rule 4 $Ca(OH)_2$ **not** $CaOH_2$

(b) ammonium carbonate

Rule 1 NH_4^+ CO_3^{2-}

Rule 2 $(NH_4^+)_2(CO_3^{2-})_1$

Rule 3 Not required

Rule 4 $(NH_4)_2CO_3$

EXERCISE 24 Write formulae for

1. sodium hydroxide
2. magnesium nitrate
3. ammonium acetate
4. aluminium phosphate
5. iron(II) carbonate
6. magnesium phosphate
7. barium sulfate
8. ammonium sulfite
9. copper(II) hydrogencarbonate
10. potassium carbonate

EXERCISE 25 Write names for

1. $AgNO_3$ 2. $Fe(OH)_2$ 3. $CaCO_3$ 4. $(NH_4)_2SO_4$ 5. $Al_2(SO_4)_3$

EXERCISE 26 Write formulae for

1. silver fluoride
2. sodium oxide
3. ammonium phosphate
4. lead(II) sulfate
5. aluminium carbonate
6. copper(II) nitrite
7. mercury(II) oxide
8. iron(II) sulfide
9. zinc hydroxide
10. lithium hydroxide

CLASSIFICATION OF ELEMENTS AND COMPOUNDS ACCORDING TO BONDING TYPE

It has been shown that the chemical bonding in elements or compounds can be classified as —

(a) *no bonding* — the noble gases.
(b) *covalent molecular* — most non-metal elements and most compounds containing only non-metals.
(c) *covalent network* — a few non-metal elements and a few compounds containing only non-metals.
Remember; C, Si, SiC and SiO_2.
(d) *ionic* — compounds of a metal with non-metals.
(e) *metallic* — metal elements.

CHAPTER 6: Chemical Equations

A chemical equation is a shorthand way of describing a chemical change. When writing an equation —

(a) *WRITE CHEMICAL FORMULAE FOR EACH SUBSTANCE,* to indicate the forms in which these substances exist.

(b) *BALANCE THE EQUATION BY PUTTING APPROPRIATE NUMBERS, CALLED COEFFICIENTS, IN FRONT OF THE FORMULAE.* The equation is balanced when the number of atoms of each element on the reactant side is the same as the number of atoms of that element on the product side. *The formulae themselves can NOT be altered.* Equations are balanced to satisfy the Law of Conservation of Mass.

(c) Often the phase of a substance is indicated by an abbreviation (in brackets) after the formula of the substance.

Abbreviation	Meaning
(g)	gas
(l)	liquid
(s)	solid
(aq)	in aqueous solution

EXAMPLE 14: Balance the equation:

$$Mg + O_2 \longrightarrow MgO$$

Oxygen is not balanced. First balance oxygen by putting a 2 before the MgO then, as the magnesium is no longer balanced, place a 2 before the Mg.

$$2Mg + O_2 \longrightarrow 2MgO$$

An equally correct balanced equation is obtained by multiplying or dividing the coefficients of a balanced equation by the same number.

e.g. $4Mg + 2O_2 \longrightarrow 4MgO$, or, $Mg + \frac{1}{2}O_2 \longrightarrow MgO$

However, usually the smallest whole numbers are used.

EXAMPLE 15: Balance the equation:

$$HCl + CaCO_3 \longrightarrow CaCl_2 + CO_2 + H_2O$$

H and Cl are not balanced, hence:

$$2HCl + CaCO_3 \longrightarrow CaCl_2 + CO_2 + H_2O$$

EXERCISE 31 Balance the following equations.

1. $Na + Cl_2 \longrightarrow NaCl$
2. $CO + O_2 \longrightarrow CO_2$
3. $Al + O_2 \longrightarrow Al_2O_3$
4. $HgO \longrightarrow Hg + O_2$
5. $Mg + N_2 \longrightarrow Mg_3N_2$
6. $P_4 + O_2 \longrightarrow P_4O_{10}$
7. $Al + F_2 \longrightarrow AlF_3$
8. $PbO + C \longrightarrow Pb + CO_2$
9. $CH_4 + O_2 \longrightarrow CO_2 + H_2O$
10. $NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$
11. $Ca + H_2O \longrightarrow Ca(OH)_2 + H_2$
12. $Fe + HCl \longrightarrow FeCl_2 + H_2$
13. $MgCO_3 + HCl \longrightarrow MgCl_2 + H_2O + CO_2$
14. $NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O$
15. $Mg + AgNO_3 \longrightarrow Mg(NO_3)_2 + Ag$

EXERCISE 32 Write names for the elements and compounds in the equations for Exercise 31.

EXAMPLE 16: Balance the equation:

$$Al + HCl \longrightarrow AlCl_3 + H_2$$

Both the H and Cl are not balanced. Balance the Cl first since H_2 contains no other elements, enabling H to be balanced last.

$$Al + 3\,HCl \longrightarrow AlCl_3 + H_2$$

Balancing the H

$$Al + 3HCl \longrightarrow AlCl_3 + \tfrac{3}{2}H_2 \qquad (\tfrac{3}{2} \times H_2 = 3H)$$

The above equation is satisfactory but the fraction can be eliminated by multiplying by 2

$$2Al + 6HCl \longrightarrow 2AlCl_3 + 3H_2$$

EXAMPLE 17: Balance the equation:

$$(NH_4)_2CO_3 + HNO_3 \longrightarrow NH_4NO_3 + H_2O + CO_2$$

NH_4^+ and NO_3^- ions appear on both sides of the equation. It is easier to balance the equation by balancing these groups, rather than dividing them into the elements of which they are made.
Neglecting these groups, C and O are balanced, but not H.

Balance the H in HNO_3.

$$(NH_4)_2CO_3 + 2HNO_3 \longrightarrow NH_4NO_3 + H_2O + CO_2$$

Simultaneously balance NH_4^+ and NO_3^-

$$(NH_4)_2CO_3 + 2HNO_3 \longrightarrow 2NH_4NO_3 + H_2O + CO_2$$

EXERCISE 33 Balance the following equations.

1. $H_2S + O_2 \longrightarrow H_2O + SO_2$
2. $Al + AgNO_3 \longrightarrow Al(NO_3)_3 + Ag$
3. $NaOH + H_3PO_4 \longrightarrow Na_3PO_4 + H_2O$
4. $KClO_3 \longrightarrow KCl + O_2$
5. $NaOH + Fe(NO_3)_2 \longrightarrow Fe(OH)_2 + NaNO_3$
6. $Fe_2O_3 + CO \longrightarrow Fe + CO_2$
7. $Cu + HNO_3 \longrightarrow Cu(NO_3)_2 + H_2O + NO_2$
8. $NH_3 + CuO \longrightarrow Cu + H_2O + N_2$
9. $Al_2(CO_3)_3 + HCl \longrightarrow AlCl_3 + H_2O + CO_2$
10. $FeS + O_2 \longrightarrow Fe_2O_3 + SO_2$

EXERCISE 34 Write names for the elements and compounds in the equations for Exercise 33.

EXAMPLE 18: Balance the equation:

hydrogen + chlorine ⟶ hydrogen chloride

$$H_2 + Cl_2 \longrightarrow HCl$$

H and Cl are unbalanced, hence:

$$H_2 + Cl_2 \longrightarrow 2HCl$$

EXAMPLE 19: Balance the equation:

Calcium hydroxide + phosphoric acid ⟶ calcium phosphate + water

$$Ca(OH)_2 + H_3PO_4 \longrightarrow Ca_3(PO_4)_2 + H_2O$$

Balance Ca

$$3Ca(OH)_2 + H_3PO_4 \longrightarrow Ca_3(PO_4)_2 + H_2O$$

Balance PO_4^{3-}

$$3Ca(OH)_2 + 2H_3PO_4 \longrightarrow Ca_3(PO_4)_2 + H_2O$$

Balance H and O (other than that in PO_4^{3-})

$$3Ca(OH)_2 + 2H_3PO_4 \longrightarrow Ca_3(PO_4)_2 + 6H_2O$$

EXERCISE 35 Write fully balanced equations for the following.

1. hydrogen + oxygen ⟶ water
2. carbon + oxygen ⟶ carbon dioxide
3. sodium + chlorine ⟶ sodium chloride
4. potassium + oxygen ⟶ potassium oxide
5. magnesium + oxygen ⟶ magnesium oxide
6. magnesium + hydrochloric acid ⟶ magnesium chloride + hydrogen
7. ammonia + water ⟶ ammonium hydroxide
8. zinc + hydrochloric acid ⟶ zinc chloride + hydrogen
9. sodium + sulfuric acid ⟶ sodium sulfate + hydrogen
10. calcium carbonate + hydrochloric acid ⟶ calcium chloride + water + carbon dioxide
11. hydrogen + nitrogen ⟶ ammonia
12. aluminium + oxygen ⟶ aluminium oxide
13. calcium oxide + water ⟶ calcium hydroxide
14. calcium hydroxide + carbon dioxide ⟶ calcium carbonate + water
15. ammonia + sulfuric acid ⟶ ammonium sulfate
16. silver nitrate + magnesium chloride ⟶ silver chloride + magnesium nitrate
17. potassium nitrate ⟶ potassium nitrite + oxygen
18. sodium hydroxide + carbon dioxide ⟶ sodium carbonate + water
19. potassium bicarbonate ⟶ potassium carbonate + water + carbon dioxide
20. zinc + silver nitrate ⟶ zinc nitrate + silver

EQUATIONS SHOW THE RATIO IN WHICH MOLECULES, ATOMS OR IONS REACT.

EXAMPLE 20: What information is given by the equation below?

$$2Na(s) + F_2(g) \longrightarrow 2NaF(s)$$

For every two atoms of sodium in a metallic lattice that react,
— one molecule of fluorine reacts
— two sodium ions and two fluoride ions in an ionic lattice are formed.

EXERCISE 36 What information is given by each of the equations below?

1. $2H_2(g) + O_2(g) \longrightarrow 2H_2O(g)$
2. $C(s) + O_2(g) \longrightarrow CO_2(g)$
3. $2KHCO_3(s) \longrightarrow K_2CO_3(s) + H_2O(g) + CO_2(g)$
4. $Zn(s) + 2AgNO_3(aq) \longrightarrow Zn(NO_3)_2(aq) + 2Ag(s)$
5. $3H_2(g) + N_2(g) \longrightarrow 2NH_3(g)$

CHAPTER 7: General Reactions

Substances with similar chemical properties will react in similar ways and therefore many chemical reactions can be classified into general reaction types. Knowledge of general reactions enables products of specific reactions to be predicted.

The most common general reaction types will be treated here.

NEUTRALIZATION

Neutralization is the term used to describe the reaction between an *acid* and a *base*. The general equations for the many different forms of acids and bases are indicated here:

ACID + BASE

1.a. ACID + METAL HYDROXIDE → SALT + WATER

e.g. $HCl + NaOH \longrightarrow NaCl + H_2O$

1.b. ACID + METAL OXIDE → SALT + WATER

e.g. $H_2SO_4 + CaO \longrightarrow CaSO_4 + H_2O$

2.a. ACID + METAL CARBONATE → SALT + WATER + CARBON DIOXIDE

e.g. $2HCl + CaCO_3 \longrightarrow CaCl_2 + H_2O + CO_2$

2.b. ACID + METAL HYDROGEN-CARBONATE → SALT + WATER + CARBON DIOXIDE

e.g. $HNO_3 + NaHCO_3 \longrightarrow NaNO_3 + H_2O + CO_2$

An ACID is a substance that produces hydrogen ions (H^+) in solution.

e.g. (i) Common acid : $HCl(aq) \longrightarrow H^+(aq) + Cl^-(aq)$

(ii) Non-metal oxide :

$$SO_2(g) + H_2O \longrightarrow H_2SO_3(aq) \rightleftharpoons 2H^+(aq) + SO_3^{2-}(aq)$$
$$SO_3(g) + H_2O \longrightarrow H_2SO_4(aq) \rightleftharpoons 2H^+(aq) + SO_4^{2-}(aq)$$
$$CO_2(g) + H_2O \longrightarrow H_2CO_3(aq) \rightleftharpoons 2H^+(aq) + CO_3^{2-}(aq)$$
$$P_4O_{10}(g) + 6H_2O \longrightarrow 4H_3PO_4(aq) \rightleftharpoons 12H^+(aq) + 4PO_4^{3-}(aq)$$

i.e. [Non-metal oxide] + [Water] ⟶ [Acid]

NOTE: 1. The use of $\rightleftharpoons$ indicates that the reaction has not gone to completion.
2. The acid in any of the general neutralization reactions above could be a non-metal oxide, in which case the salt formed is due to the acid produced when the non-metal oxide dissolves in water.
e.g. $SO_2 + 2NaOH \longrightarrow Na_2SO_3 + H_2O$ (equation type (1.a.))
3. Sulfur dioxide is a major pollutant causing "acid rain".

The following acids should be memorised:

NAME OF ACID	FORMULA	USE
Hydrochloric acid	HCl	Cleans bricks and iron
Sulfuric acid	H_2SO_4	Car batteries, make fertilizers
Nitric acid	HNO_3	Make fertilizers, explosives
Phosphoric acid	H_3PO_4	Make fertilizers
Acetic acid	CH_3COOH	Vinegar
Carbonic acid	H_2CO_3	—
Sulfurous acid	H_2SO_3	—

Acids taste sour, feel rough and are corrosive.

A BASE (or alkali) is a substance that produces hydroxide ions (OH^-) in solution.

e.g. (i) Metal hydroxide : $NaOH(aq) \longrightarrow Na^+(aq) + OH^-(aq)$

(ii) Metal oxide: $CaO(s) + H_2O \longrightarrow Ca(OH)_2(aq) \longrightarrow Ca^{2+}(aq) + 2OH^-(aq)$

(iii) Metal carbonate: $Na_2CO_3(s) + H_2O \rightleftharpoons 2Na^+(aq) + OH^-(aq) + HCO_3^-(aq)$

(iv) Metal hydrogencarbonate: $KHCO_3(s) + H_2O \rightleftharpoons K^+(aq) + OH^-(aq) + H_2CO_3(aq)$

NOTE:1. Although they contain no metal ions, both ammonium carbonate, $(NH_4)_2CO_3$, and ammonium hydrogencarbonate, NH_4HCO_3, also behave as bases.

2. Ammonia solution is a common base. Its property of being a base can be seen in the following:

$$NH_3(g) + H_2O \rightleftharpoons NH_4OH(aq) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

The following bases should be memorised:

COMMON NAME OF BASE	FORMULA	USE
Caustic soda	NaOH	Make soap, clean drains
Milk of magnesia	$Mg(OH)_2$	Relieve an upset stomach
Ammonia	NH_3	Make fertilisers, explosives
Washing soda	Na_2CO_3	Make glass, soften water
Baking soda	$NaHCO_3$	Bake cakes
Quicklime	CaO	Make lime mortar, plaster
Slaked lime	$Ca(OH)_2$	

Bases taste bitter, feel slippery and are corrosive. They are used in household cleaners, dishwasher powder and oven cleaners to remove grease and fats.

EXERCISE 37: ACIDS AND BASES

Write the formula or name of each of the following substances and then classify it as an acid or a base:

1. Carbon dioxide
2. Magnesium oxide
3. Li_2O
4. $Ca(HCO_3)_2$
5. Barium carbonate
6. KOH
7. H_3PO_4
8. Sulfur trioxide
9. NH_3
10. Nitrogen dioxide

A SALT is produced by the combination of the positive ion from the base and the negative ion from the acid.

For example, NaCl(aq) is the salt formed when HCl(aq) is reacted with NaOH(aq).

WATER is produced in neutralization through the combination of the hydrogen ion, $H^+(aq)$, from the acid and the hydroxide ion, $OH^-(aq)$, from the base.

EXAMPLE 21: Write the balanced equation for the reaction between dilute hydrochloric acid and copper(II) oxide solid.

ANSWER: By equation type (1.b.)

$$2HCl(aq) + CuO(s) \longrightarrow CuCl_2(aq) + H_2O(l)$$

EXERCISE 38: NEUTRALIZATION

Write the balanced equations for the reactions produced when the following substances are mixed:

1. Dilute sulfuric acid + potassium hydroxide solution.
2. Dilute hydrochloric acid + calcium hydroxide.
3. Dilute hydrochloric acid + magnesium oxide.
4. Sulfur dioxide gas + sodium hydroxide solution.
5. Calcium carbonate + dilute nitric acid.
6. Acetic acid solution + iron(II) oxide.
7. Aluminium hydrogencarbonate + sulfuric acid solution.
8. Aluminium oxide + dilute hydrochloric acid.
9. Calcium hydroxide + phosphoric acid solution.
10. Potassium hydroxide solution + sulfur trioxide gas.
11. Sulfurous acid solution + iron(III) hydrogencarbonate solution.
12. Barium hydroxide solution + phosphoric acid solution.
13. Nickel carbonate + dilute sulfuric acid.
14. Tin(II) hydrogencarbonate + dilute hydrochloric acid.
15. Dilute acetic acid + zinc hydroxide.
16. Ammonia solution + dilute nitric acid.
17. Carbonic acid + sodium hydroxide solution.
18. Tetraphosphorus decaoxide + potassium hydroxide solution.
19. Ammonium hydrogencarbonate solution + dilute hydrochloric acid
20. Dilute nitric acid + iron(III) hydroxide.

INDICATORS

An INDICATOR is a substance, often a plant dye, which has a charactertistic but different colour in acidic and basic solutions.

INDICATOR	COLOUR IN ACID	COLOUR IN BASE
Litmus	Red	Blue
Methyl orange	Red	Yellow
Phenolphthalein	Colourless	Purple

EXERCISE 39: INDICATORS

State the colour of litmus, methyl orange and phenolphthalein in solutions of each of the substances listed in Exercise 37.